Active Learning for Science

AMLAB Seminar

David Holzmüller (INRIA Paris)

February 27, 2025

AL4Science

Coauthors



Viktor Zaverkin



Daniel Musekamp

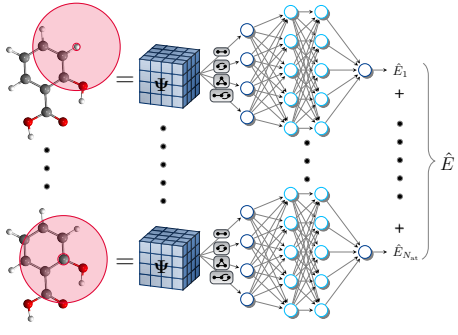


Ingo Steinwart

. . .

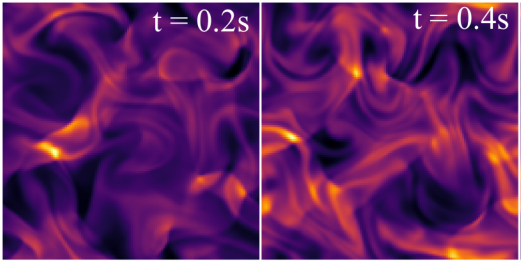
Learning surrogate models via regression

(positions, atom types) $\mapsto$ energy



Source: Zaverkin and Kästner (2020)

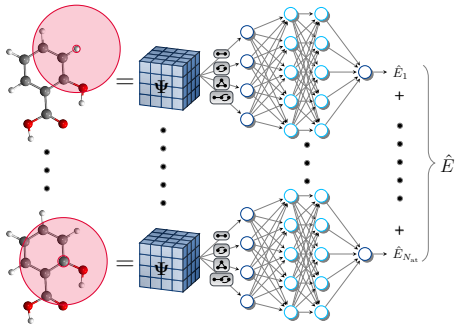
(grid, PDE parameters) $\mapsto$ grid



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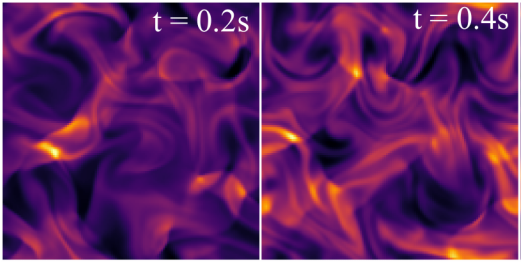
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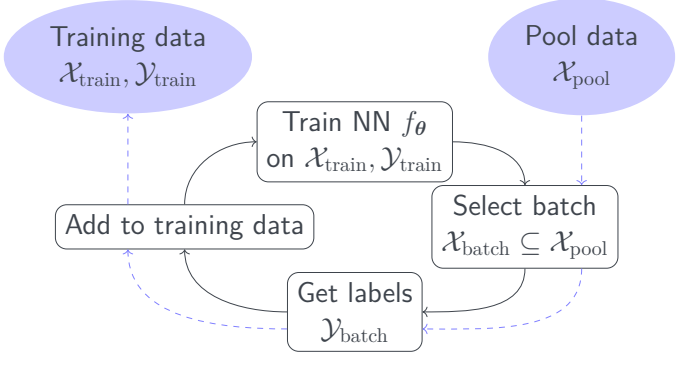


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Active Learning: Where should we query the simulator?

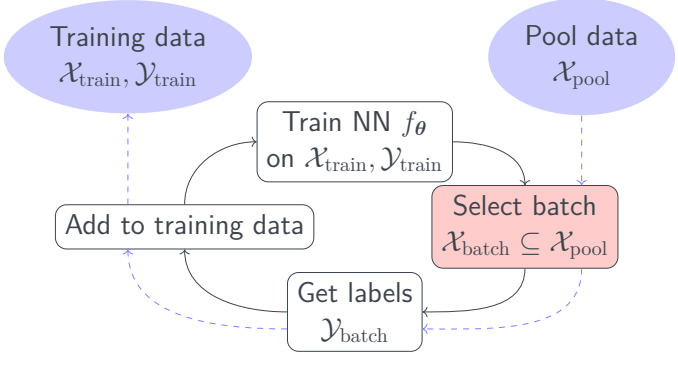
Pool-based active learning

Active learning loop



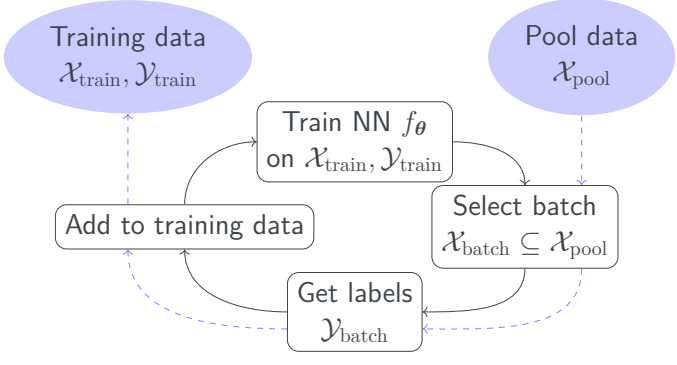
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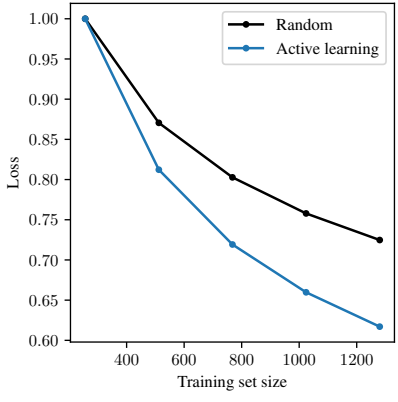


Pool-based active learning

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Desired result



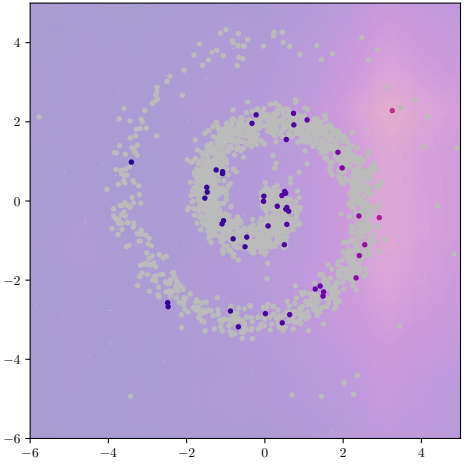
Tabular data

Material type	Composition ratio	Cooling time	...	Energy efficiency
A	0.4	2.5	...	20%
B	0.6	3.5	...	22%
⋮	⋮	⋮		⋮

- Heterogeneous features. Here: Learning functions vector $\mapsto$ scalar
- We look at NNs although other good methods exist
- Allows for cheap and diverse benchmarks
- My current research topic

David Holzmüller, Viktor Zaverkin, Johannes Kästner, and Ingo Steinwart. A framework and benchmark for deep batch active learning for regression.
Journal of Machine Learning Research, 24(164), 2023

Example: Linear regression with feature map ϕ



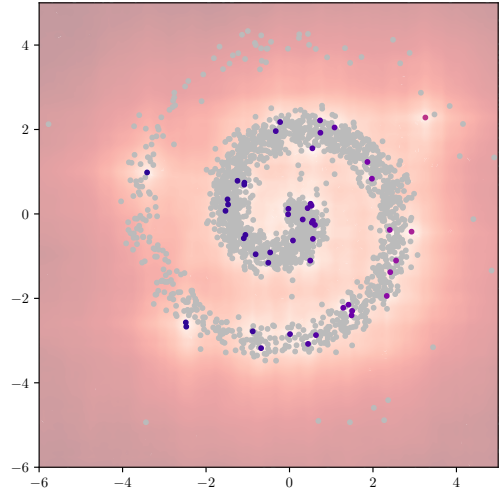
Kernel regression with $k(x, x') = \phi(x)^\top \phi(x') \Leftrightarrow$ linear regression with feature map ϕ

Active learning: Desiderata (Wu, 2018, H. et al., 2023)

- **Uncertainty**
  - GP uncertainty (*epistemic*)
    $k(x, X)(k(X, X)^{-1} + \lambda I)^{-1} k(X, x)$

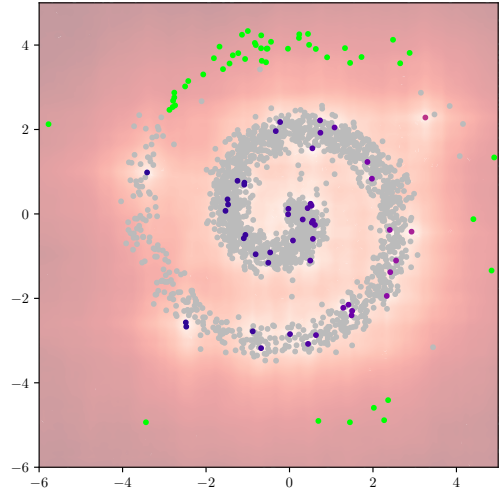
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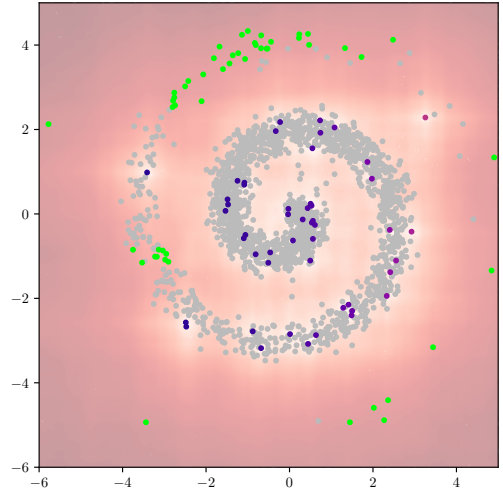
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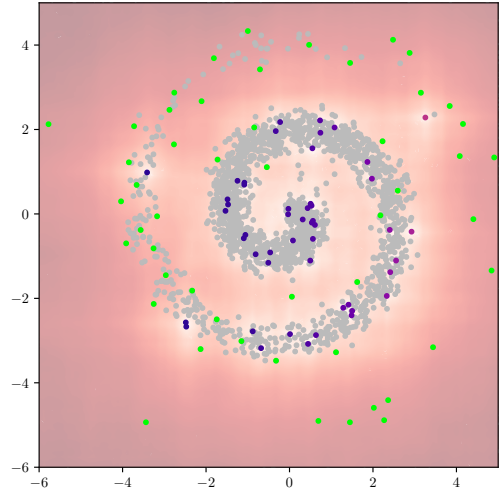
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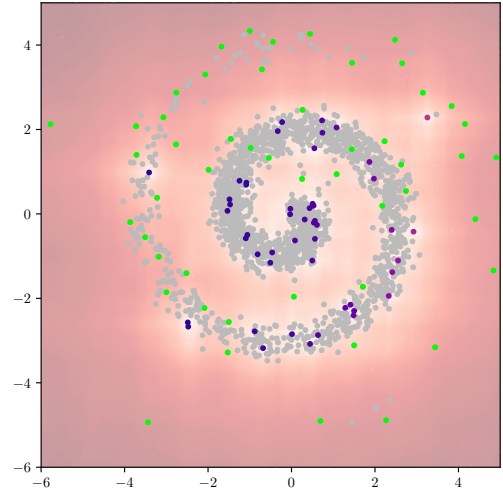
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  - GP: Update X after each point



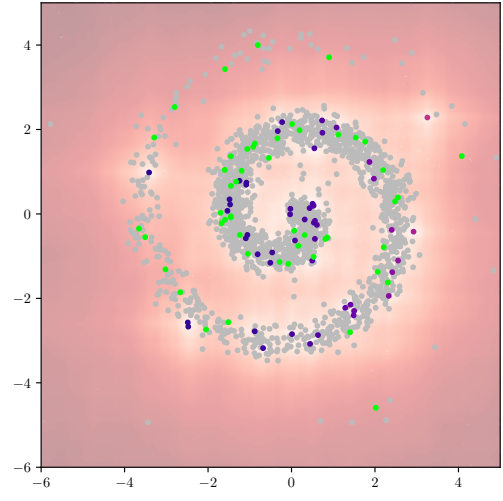
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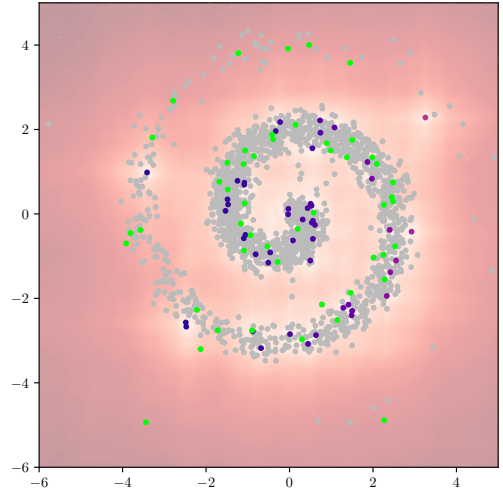
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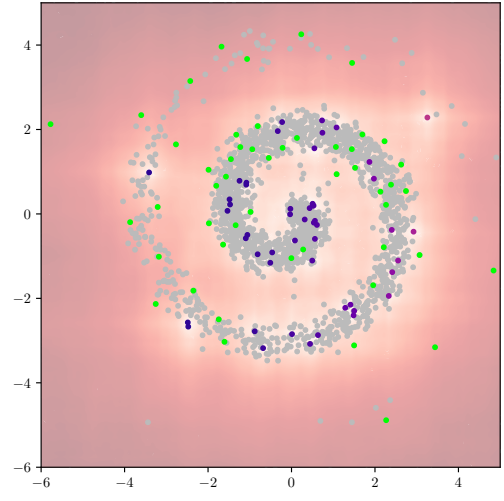
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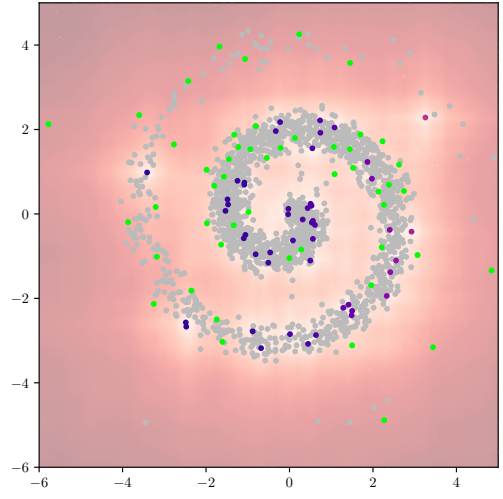
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  - LCMD clustering (H. et al., 2023)



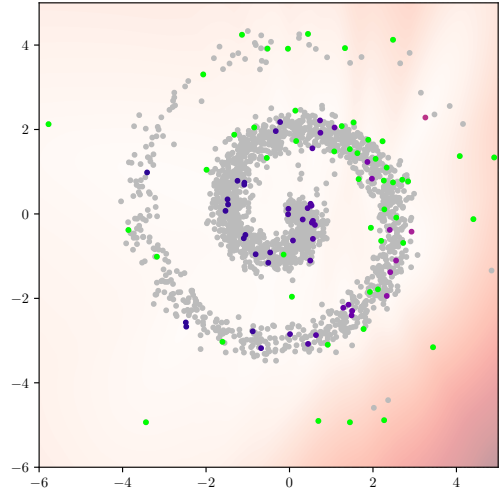
Features

- Fixed feature map $\phi \rightsquigarrow$ sampling completely independent of labels



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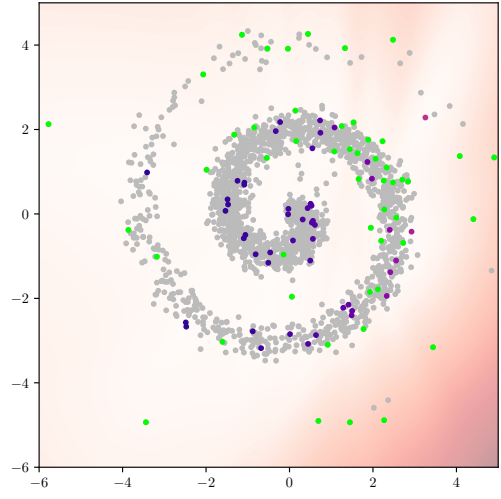
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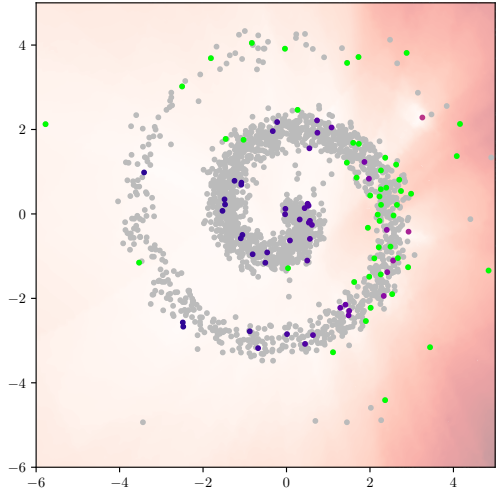
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- NN gradient features (H. et al., 2023)

$$\phi(x) \coloneqq U \nabla_\theta f_\theta(x)$$

with random sketching matrix U



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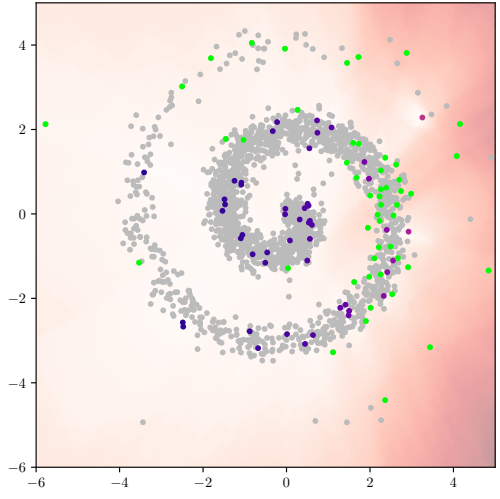
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- Vectors of centered ensemble predictions (Kirsch, 2023)

Benchmark results (H. et al., 2023)

Benchmark: 15 tabular regression datasets, 30K–200K pool points

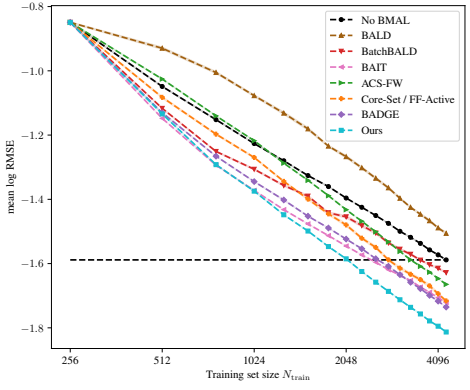
`github.com/dholzmueller/bmdal_reg`

Benchmark results (H. et al., 2023)

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Comparison to literature

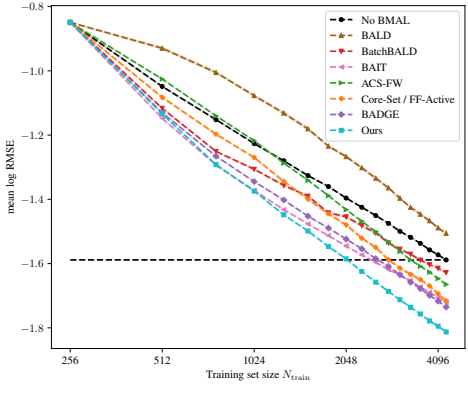


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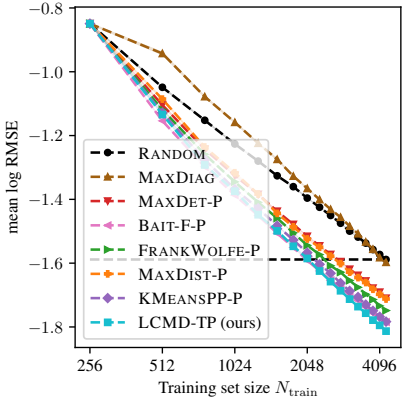
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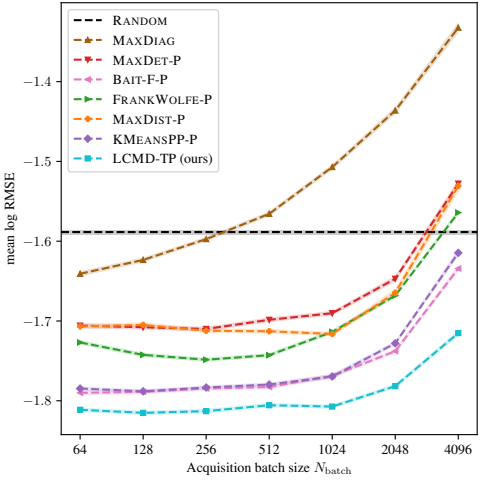
Comparison to literature



With gradient features



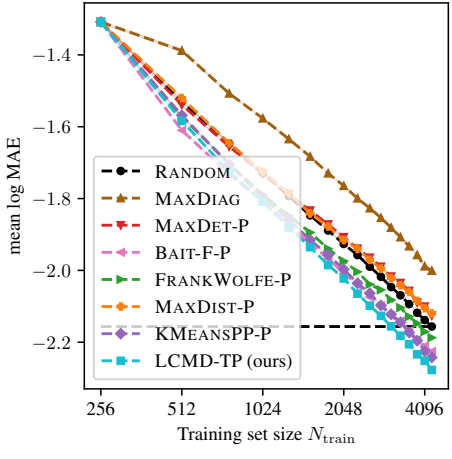
Four acquisition steps are enough



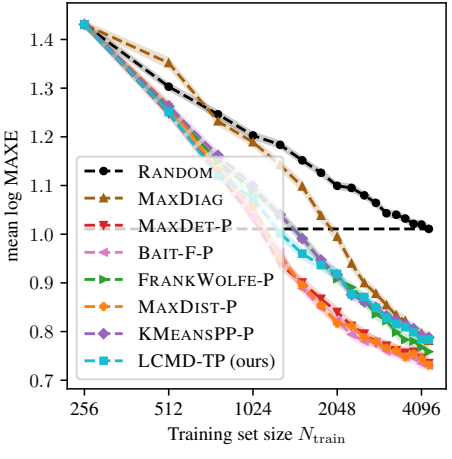
Speed: $\approx$ 1s per step for selection

Active learning trades off easy for hard examples

Mean absolute error



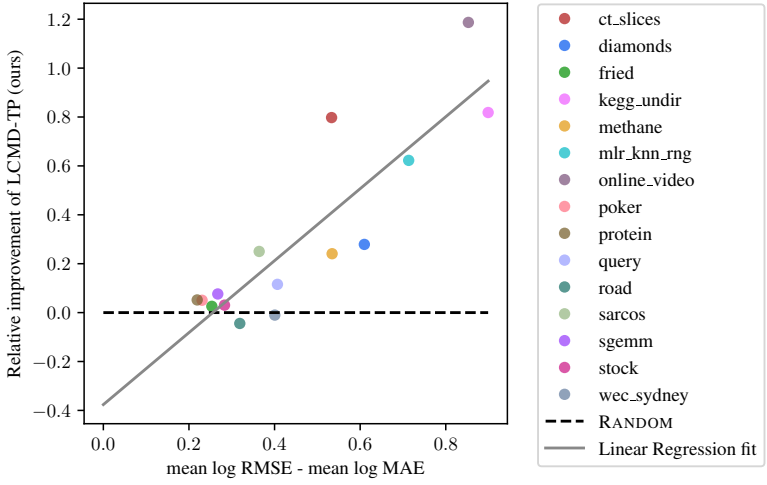
Maximum error



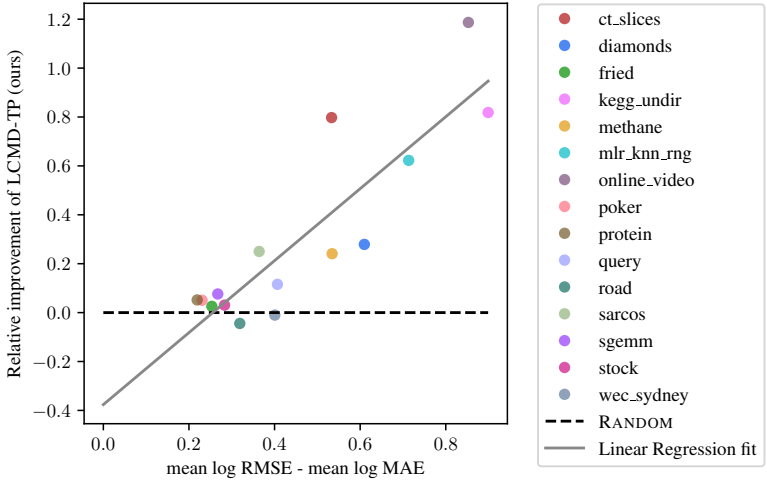
→ Active learning is more useful for tail-sensitive metrics

When does active learning help?

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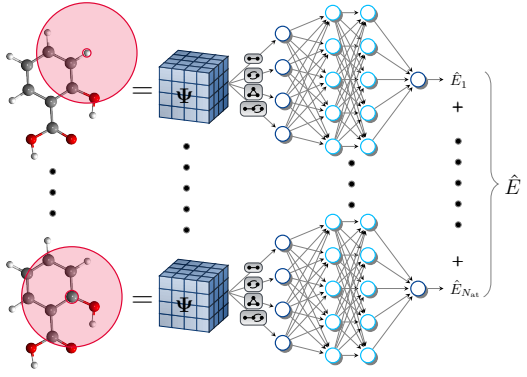


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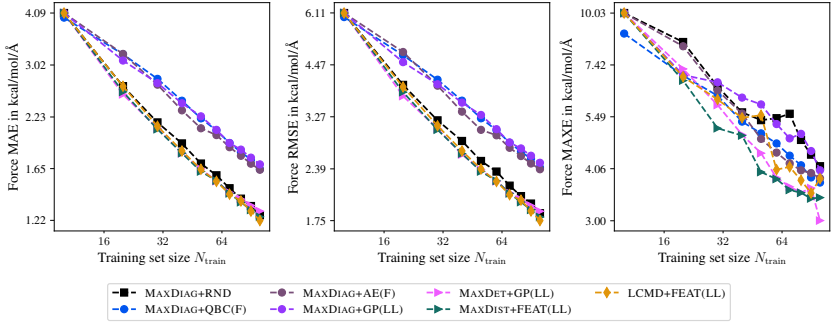
Active learning helps for heterogeneous error distributions.

Active learning for interatomic potentials (Zaverkin and Kästner 2021, Zaverkin, H., et al. 2022)



- Model "atomic energies", but only their sum is known $\rightsquigarrow$ sum up per-atom features
- Train with forces $(-\nabla E(x))$ $\rightsquigarrow$ ignore for active learning

Performance on MD17 aspirin (Zaverkin, H., et al., 2023)



- AL not very helpful on QM9 and LMNTO, very helpful on TiO2
- Not much difference between last-layer and gradient features

What if there is no pool set?

In general: Query-synthesis active learning
Here: Care about the Boltzmann distribution

$$p(x) = \exp\left(-\frac{E(x)}{k_B T}\right)/Z$$

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Procedure in Zaverkin et al. (2024) and others:

- Initial data: small random distortions of initial structure
- Loop
  - Compute energies and forces on new data using ground-truth (e.g., DFT)
  - Train model
  - Run molecular dynamics on model to generate a pool set
  - Stop when uncertainty exceeds threshold
  - Select new configurations using active learning

Bootstrapping using calibrated uncertainties

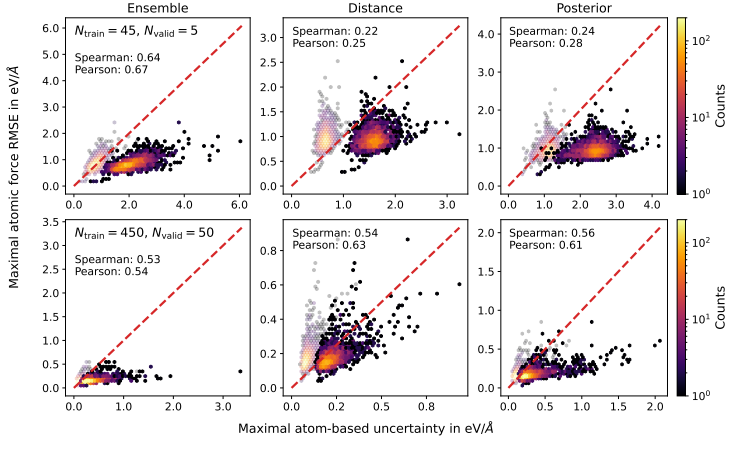


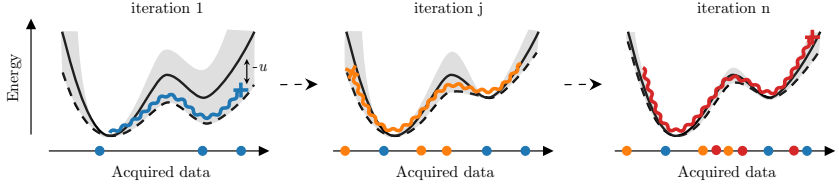
Figure: Uncertainties re-scaled to have 95% coverage (transparent: 50%)

→ split conformal prediction to obtain pessimistic uncertainties

Uncertainty biasing for exploring rare events

Approaches towards rare event exploration

- Increase temperature/pressure $\rightsquigarrow$ might break at unknown point
- Metadynamics $\rightsquigarrow$ need good collective variables
- **Here: Uncertainty biasing** (Kulichenko et al., 2023; van der Oord et al., 2023; Zaverkin et al., 2024)



Uncertainty biasing can jump between meta-stable states

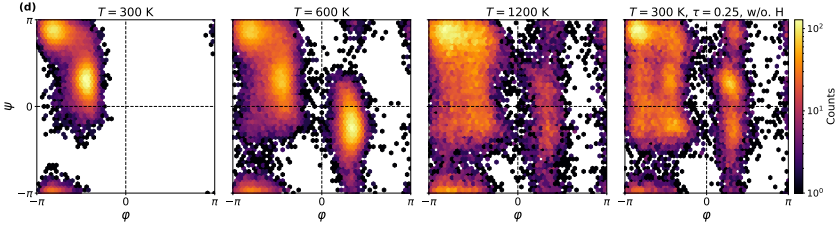
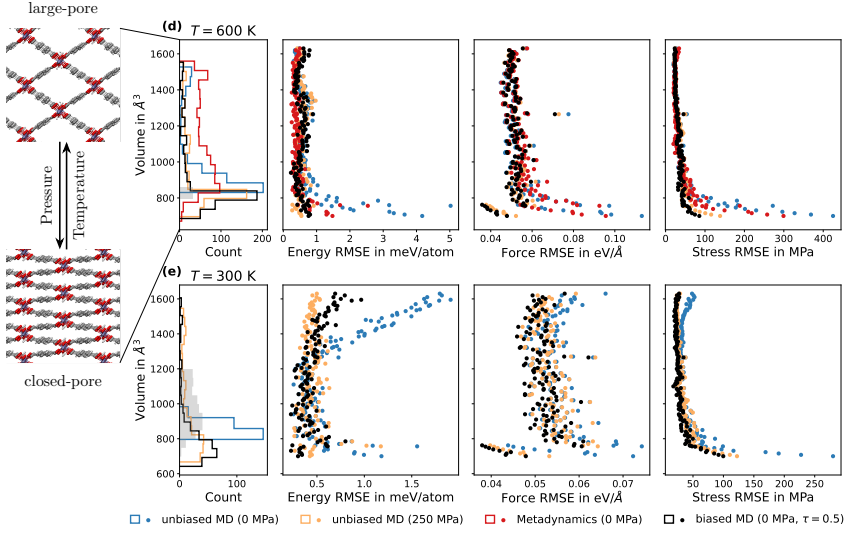


Figure: Exploration of two angles within alanine dipeptide.

Caveat: Many details to get right (don't bias H atoms, adaptive normalization, . . .)

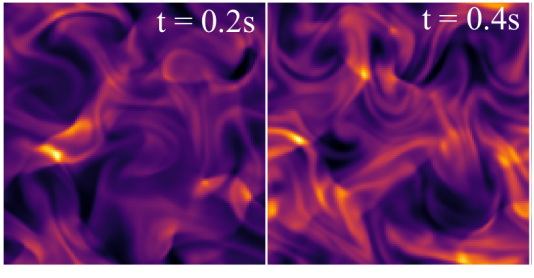
Exploring two metastable states in a metal-organic framework



Benchmark: Active Learning for Neural PDE Solvers

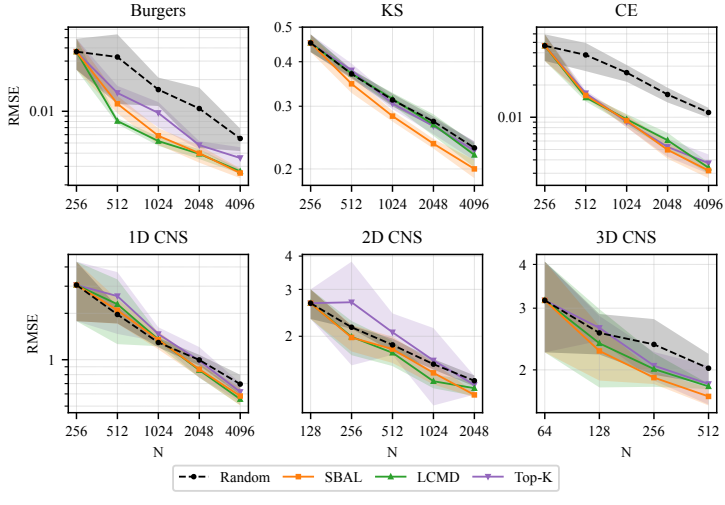
Neural PDE solvers

- Goal: Learn mapping (grid, PDE parameters) $\mapsto$ grid
- Neural operators: U-Net, Fourier Neural Operator, ...



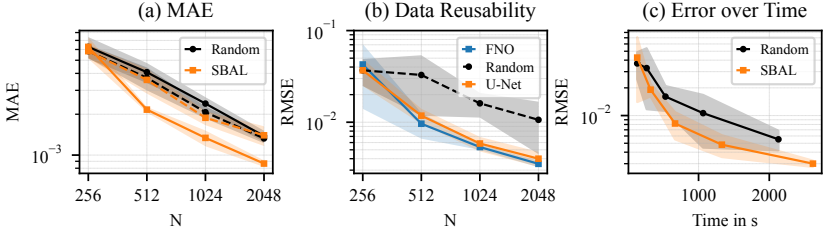
Introduce an extensible benchmark in Musekamp, Kalimuthu, H., et al. (2024) ($\rightsquigarrow$ ICLR 2025)

Performance on PDE problems



SBAL (Kirsch et al., 2023): $P(\text{select } x) \propto \text{uncertainty}(x)$

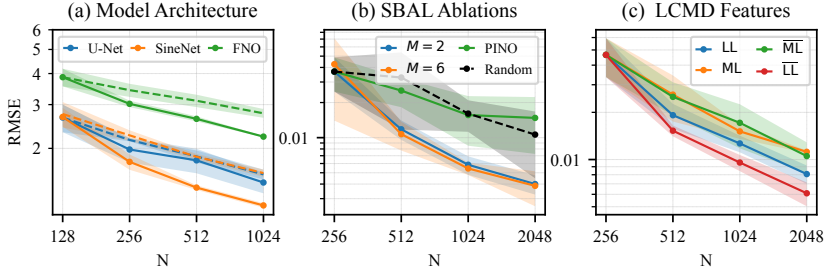
Reusability and speed



→ Can re-use data generated from different models

→ Can achieve wall-clock speed-ups even for cheap simulators

Ablations



→ 2 ensemble members are enough

→ Spatial feature averaging introduces invariances

Conclusion

Convenient batch active learning for NN regression possible:

- post-hoc, fast, does not require ensembles
- select uncertain, diverse, and representative samples
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- Extensions to richer selection spaces (multi-fidelity simulations etc.)?

Literature I

V. Zaverkin and J. Kästner. Gaussian Moments as Physically Inspired Molecular Descriptors for Accurate and Scalable Machine Learning Potentials. *Journal of Chemical Theory and Computation*, 16(8):5410–5421, August 2020. ISSN 1549-9618, 1549-9626. doi: 10.1021/acs.jctc.0c00347. URL https://pubs.acs.org/doi/10.1021/acs.jctc.0c00347.

Daniel Musekamp, Marimuthu Kalimuthu, David Holzmüller, Makoto Takamoto, and Mathias Niepert. Active Learning for Neural PDE Solvers, August 2024. URL http://arxiv.org/abs/2408.01536. arXiv:2408.01536 [cs].

David Holzmüller, Viktor Zaverkin, Johannes Kästner, and Ingo Steinwart. A framework and benchmark for deep batch active learning for regression. *Journal of Machine Learning Research*, 24 (164), 2023.

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Viktor Zaverkin and Johannes Kästner. Exploration of transferable and uniformly accurate neural network interatomic potentials using optimal experimental design. *Machine Learning: Science and Technology*, 2(3):035009, 2021.

Literature II

Viktor Zaverkin, David Holzmüller, Ingo Steinwart, and Johannes Kästner. Exploring chemical and conformational spaces by batch mode deep active learning. *Digital Discovery*, 1(5):605–620, 2022. Publisher: Royal Society of Chemistry.

Viktor Zaverkin, David Holzmüller, Henrik Christiansen, Federico Errica, Francesco Alesiani, Makoto Takamoto, Mathias Niepert, and Johannes Kästner. Uncertainty-biased molecular dynamics for learning uniformly accurate interatomic potentials. *npj Computational Materials*, 10(1):83, 2024. URL https://www.nature.com/articles/s41524-024-01254-1. Publisher: Nature Publishing Group UK London.

Maksim Kulichenko, Kipton Barros, Nicholas Lubbers, Ying Wai Li, Richard Messerly, Sergei Tretiak, Justin S. Smith, and Benjamin Nebgen. Uncertainty-driven dynamics for active learning of interatomic potentials. *Nature Computational Science*, 3(3):230–239, 2023. URL https://www.nature.com/articles/s43588-023-00406-5. Publisher: Nature Publishing Group US New York.

Cas van der Oord, Matthias Sachs, Dávid Péter Kovács, Christoph Ortner, and Gábor Csányi. Hyperactive learning for data-driven interatomic potentials. *npj Computational Materials*, 9(1):168, 2023. URL https://www.nature.com/articles/s41524-023-01104-6. Publisher: Nature Publishing Group UK London.

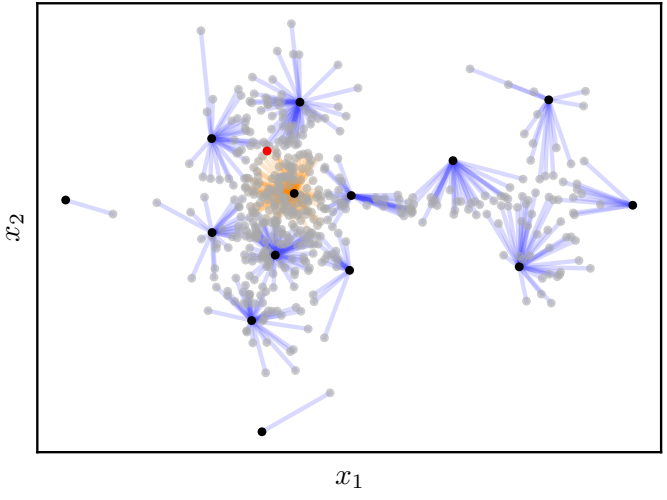
Literature III

Andreas Kirsch, Sebastian Farquhar, Parmida Atighehchian, Andrew Jesson, Frédéric Branchaud-Charron, and Yarin Gal. Stochastic Batch Acquisition: A Simple Baseline for Deep Active Learning. *Transactions on Machine Learning Research*, 2023. URL https://openreview.net/forum?id=vcHwQyNBjW.

Jordan Ash, Surbhi Goel, Akshay Krishnamurthy, and Sham Kakade. Gone fishing: Neural active learning with Fisher embeddings. In *Neural Information Processing Systems*, 2021.

What does LCMD do?

LCMD = select maximum-distance point within the "largest" cluster



Error quantiles

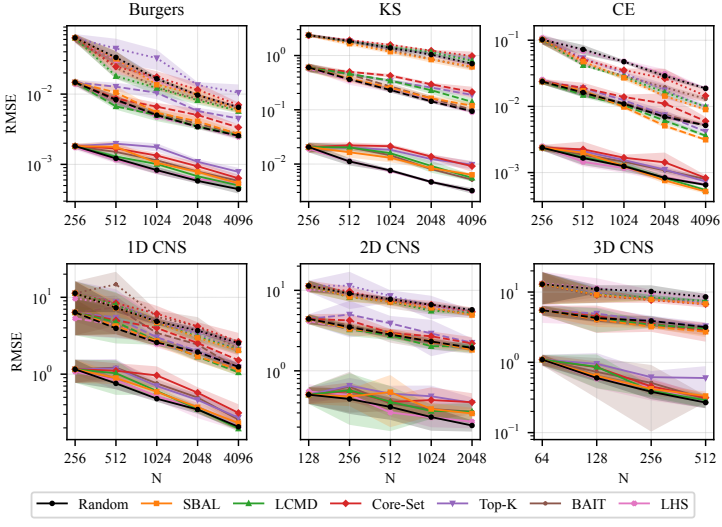
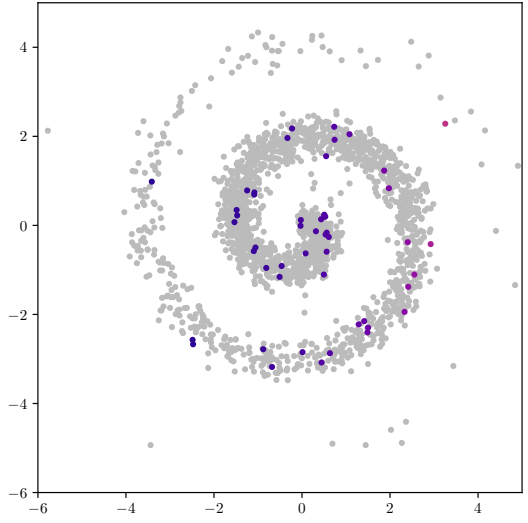


Figure: The 50%, 95%, and 99% error quantiles of the error distributions.

Example setting

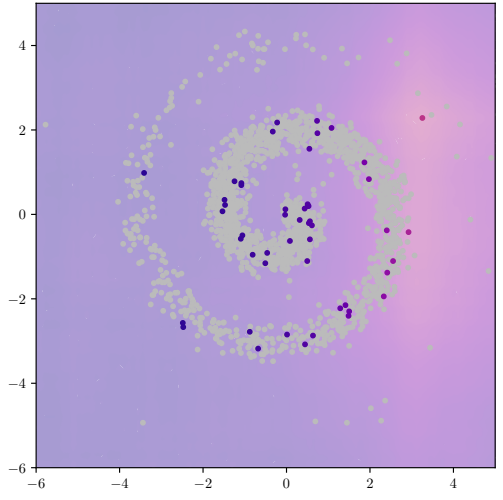


Step 1: Choose a feature map

- Choose feature map $\phi : \mathbb{R}^2 \to \mathbb{R}^p$ or kernel $k(\boldsymbol{x}, \boldsymbol{x}') = \langle \phi(\boldsymbol{x}), \phi(\boldsymbol{x}') \rangle$
- Bayesian linear regression: $y_i = \phi(\boldsymbol{x}_i)^\top \boldsymbol{\theta} + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$
- Prior: $\boldsymbol{\theta} \sim \mathcal{N}(0, \boldsymbol{I})$
- Posterior mean: $\hat{y} = \phi(\boldsymbol{x})^\top \boldsymbol{\theta}^*$, $\boldsymbol{\theta}^* = (\phi(\boldsymbol{X})^\top \phi(\boldsymbol{X}) + \sigma^2 \boldsymbol{I})^{-1} \phi(\boldsymbol{X})^\top \boldsymbol{y}$.

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Step 2: Transform to get uncertainty

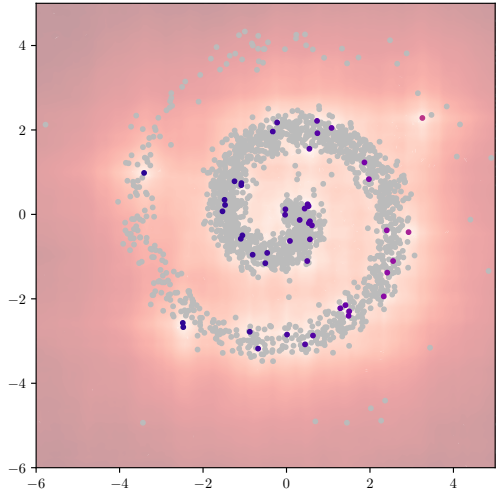
- Posterior predictive uncertainty given by transformed kernel:

$$
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& k_{\to \mathrm{post}(\mathcal{D}_{\mathrm{train}})}(\boldsymbol{x}, \boldsymbol{x}) \\
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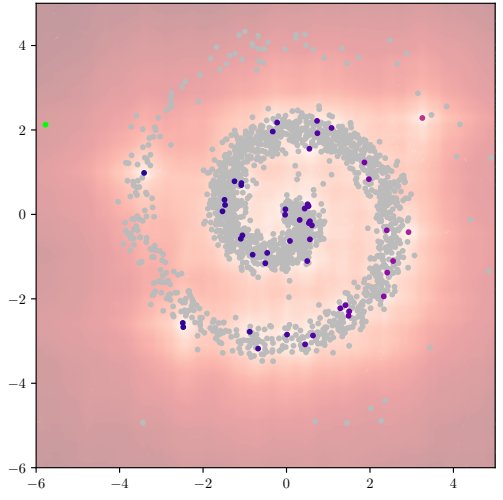


Step 3: Use uncertainty to select points

Maximum uncertainty

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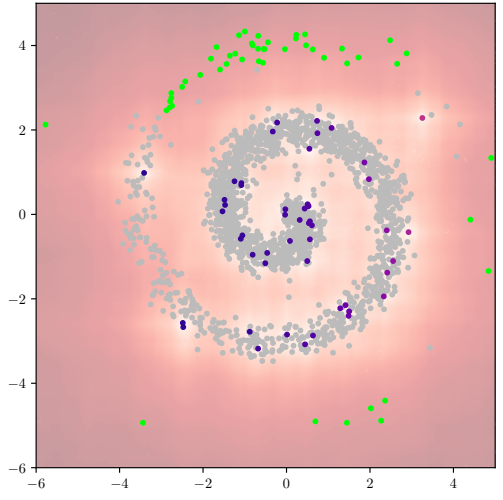


Batch setting

- Pick multiple points to label
  $\rightsquigarrow$ parallel labeling & less re-training
- Choose by uncertainty

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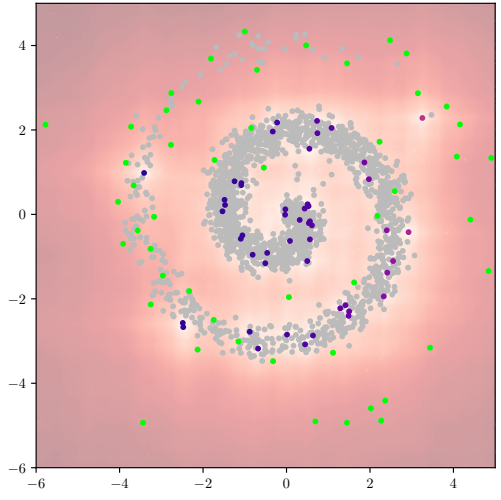


Diversity with MaxDet

- Linear model uncertainty does not depend on labels!
- Can update uncertainty after each point without knowing labels!
- We call this $\textsc{MaxDet}$, because it maximizes $\det(k(\boldsymbol{X}, \boldsymbol{X}) + \sigma^2 \boldsymbol{I})$
- Efficient implementation in $O(N_{\text{batch}}^2 N_{\text{pool}})$
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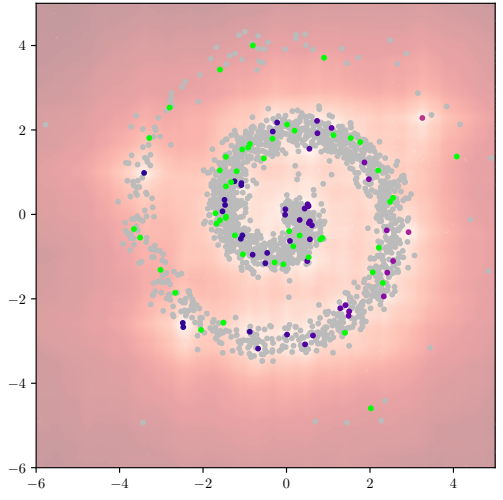


Representativity with BAIT (Ash et al., 2021)

- MaxDet does not consider which regions of space are more interesting to us
- Can deteriorate with larger pool set
- Instead, pick point that most reduces average variance on pool set
- Scales quadratically with feature dimension

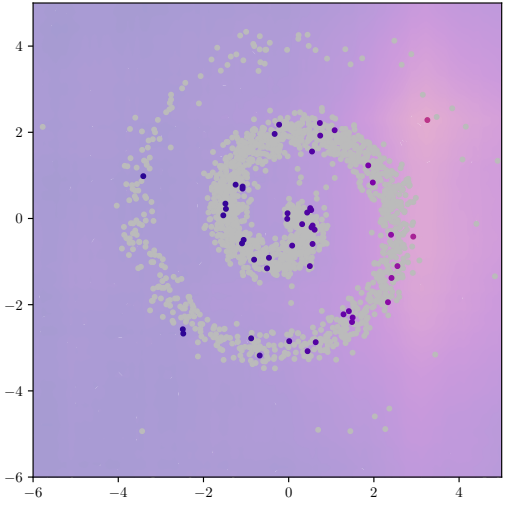
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Neural Networks

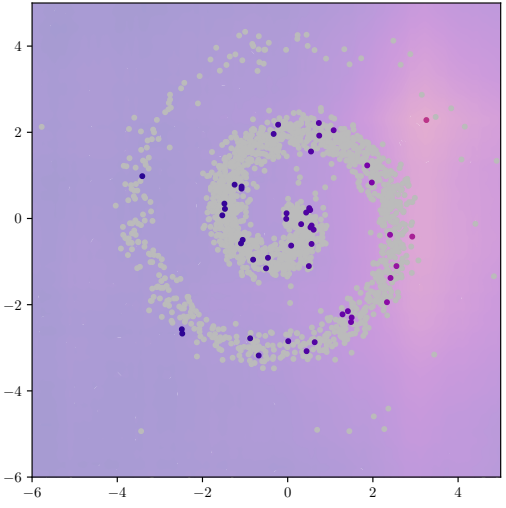
Random Fourier features



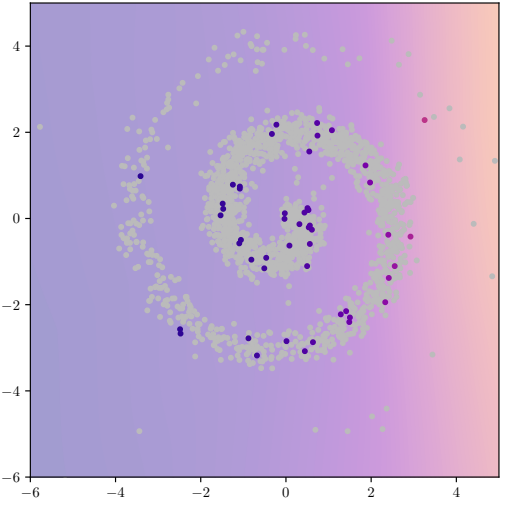
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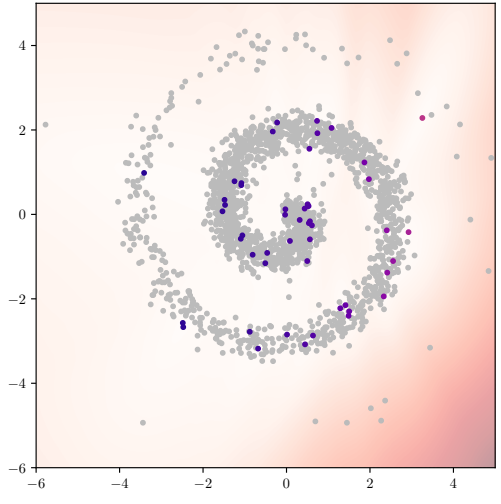


Last-layer feature map

- NN $=$ linear model with learnable feature map: $f_{\boldsymbol{\theta}}(\boldsymbol{x}) = \phi_{\boldsymbol{\theta}_{<L}}(\boldsymbol{x})^\top \boldsymbol{\theta}_L$
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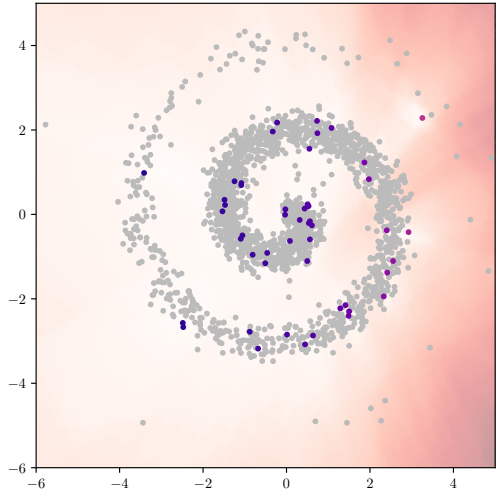


Sketched gradient feature map (new!)

- Linearize NN in parameters:
  $f_{\tilde{\boldsymbol{\theta}}}(\boldsymbol{x}) \approx f_{\boldsymbol{\theta}}(\boldsymbol{x}) + (\nabla_{\boldsymbol{\theta}} f_{\boldsymbol{\theta}}(\boldsymbol{x}))^\top (\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta})$
- Use gradient as feature map:
  $\phi_{\text{grad}}(\boldsymbol{x}) \coloneqq \nabla_{\boldsymbol{\theta}} f_{\boldsymbol{\theta}}(\boldsymbol{x})$
- Also known as empirical neural tangent kernel (NTK)
- Sketching: Use random matrix $\boldsymbol{U}$ to reduce feature dimension:
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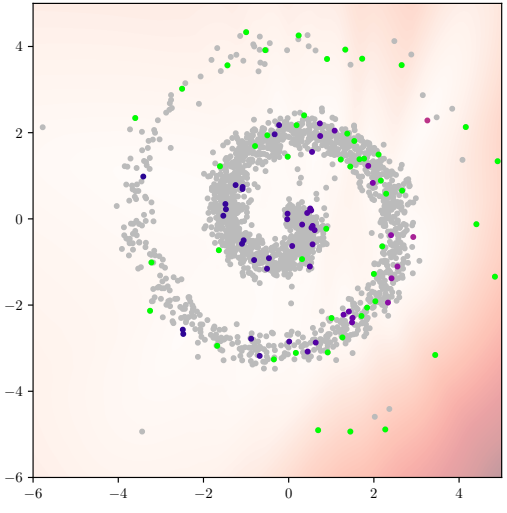
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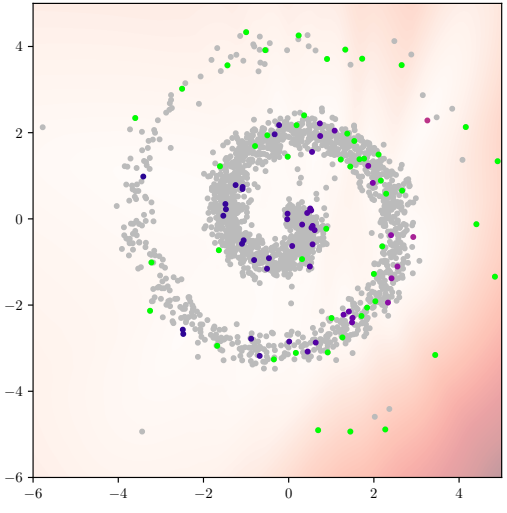
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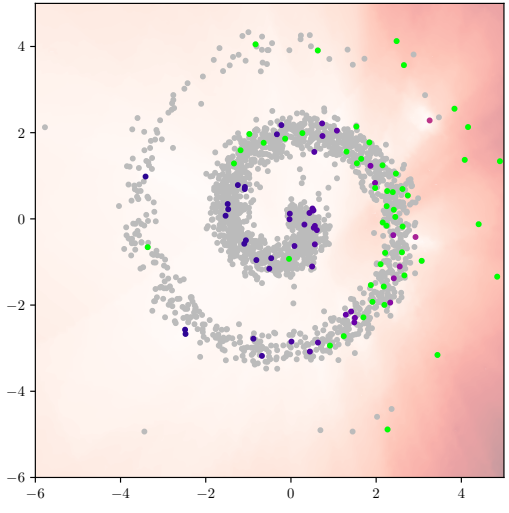
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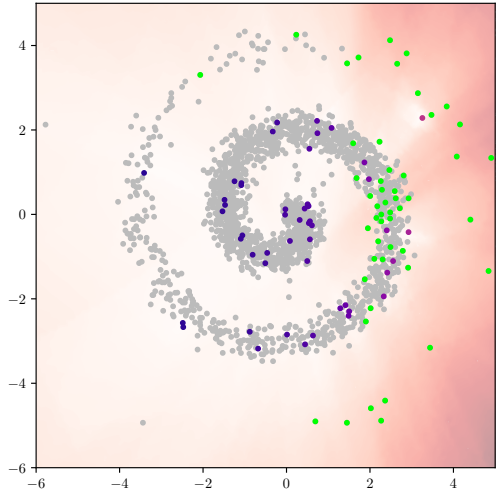


Distance-based active learning: MaxDist

- Use feature space distance $\min_i \|\phi(\boldsymbol{x}) - \phi(\boldsymbol{x}_i)\|$ as uncertainty
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- Well-known in many communities, we call it MaxDist
- Very efficient!

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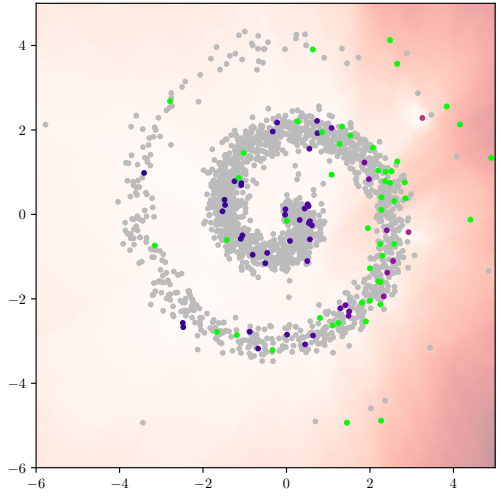


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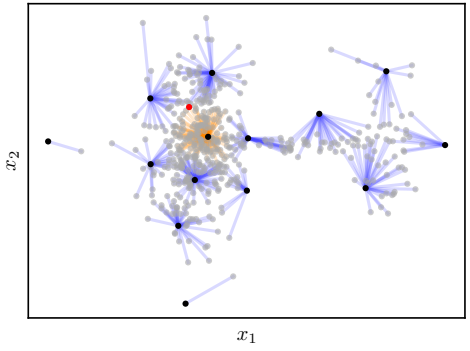
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